The 5532 OpAmplifier (2)

Part 2: construction, bridged operation and test results

By Douglas Self (UK) and Ton Giesberts (Elektor Labs)

In this second and closing instalment we get real with two times 32 NE5532 opamps paralleled to form a high-end audio power amplifier with eminent specifications in terms of distortion and general sonic performance. There are also challenges for you: bridged operation for higher output power, and modifying the amplifier for 4-ohm operation.

If you thought that paralleling a few dozen NE5532 opamps is a curious way of designing a high-end audio amplifier and typical of Elektor's off-the-beaten track approach to electronics you are probably right. Last month's design considerations did not fail to trigger responses from you, our readership, in particular from all and sundry aspiring or even claiming to be a high-end audio designer. This month we get real by building the 5532 OpAmplifier project and putting it through its paces.

Construction — amp board

The amplifier proper is built on a double-sided through plated circuit board of which the component overlay is shown in **Figure 1**. Two of these boards are required for a stereo amplifier. Boards supplied by Elektor come with through plating, a solder mask and holes predrilled. As such they are the best guarantee for success in replicating the project. While on the subject of quality, you are looking at an expensive, high-end amplifier. If you use mishmash components and *ditto* assembly methods and tools, you'll get mishmash results. More on selecting the best NE5532 brand to use in the **inset**.

Construction on the 205 × 84 mm board should not present problems as only standard through-hole parts are used. A properly functioning amp can be expected if you work with care and precision, like Ton Giesberts of Elektor Audio Labs who designed, built and tested all the boards pictured. Some notes and *caveats* deserve mentioning, however.

Most resistors are mounted vertically. Use a uniform method of bending the long terminal twice to obtain right angles. Semicircles indicate where the resistors sit on the board. Where rectangles are printed, the resistor is mounted flat on the board.

Although it's possible to solder all opamps directly on the board, you do so at a risk (see part 1), hence the prototype was built using turned-pin 8-way DIL sockets for all opamps. It is essential to use premium quality sockets here — don't be tempted to use cheap ones with dodgy spring contacts; it is false economy.

The tallest components on the board are capacitors C2, coil L1, relay RE1 and the screw terminal blocks for the loudspeaker and supply connections. The coil consists of 10 turns of 1 mm diameter (AWG18) enamelled copper wire (ECW) with an internal diameter of 20 mm. The coil windings should be spread evenly to obtain an overall length to suit the footprint on the board. Fit the coil first, then R106 at a height of 5-10 mm above the board surface and not touching L1 anywhere.

Although printed on the silk screen PCB overlay, **capacitors C24, C25, C26 and C27 are not fitted**. Instead, a single 1,000 µF, 63 V electrolytic capacitor is used, its terminals being inserted directly into terminal blocks K16/K17 (observe the polarity and provide the leads with sleeving for insulation). This should be a high quality electrolytic capacitor with low ESR. The modification proved necessary to get rid of unexpected distortion levels occurring at about 20 kHz in the initial design and was found to totally cure the problem. The amplifier specifications printed last month apply to the capacitor-modified version only.

The four corners of the amplifier board have holes for PCB standoffs — we used 10 mm tall ones.

The finished amplifier board should be given a thorough visual inspection before taking into use. Did you get all the polarised components' orientations right? Are all solder joints beyond reproach? It often helps to have a friend look at the board. **Figure 1** once again shows the amp board for your reference. See how close you can get to this level of sophistication in building high-end audio circuitry.

Construction — PSU board

This is also a classic board with nothing but standard components, hence should be easy to assemble and get working. It's just the heatsinking of the voltage regulators that requires some mechanical work.

The two bridge rectifiers B1 and B2 should be fitted with 2 mm thick 70 x 35 mm aluminium plates bolted directly to the flat sides. The two voltage regulators IC1 and IC2 are secured to a single black, finned, extruded aluminium heatsink of which the mounting outline is shown on the component overlay. The heatsink is secured to the board with three M4 screws or bolts, for which holes need to be drilled and tapped in the underside. Venting holes are provided in the PCB area under the heatsink

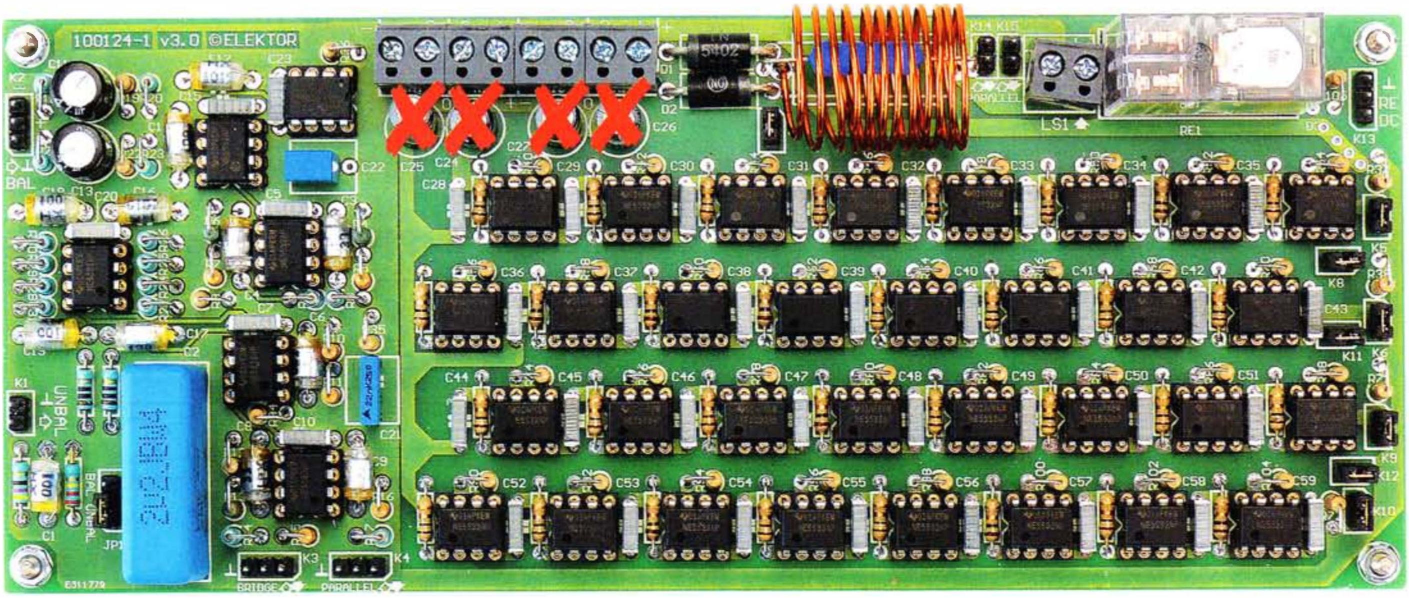


Figure 1. Perfect sound from perfect construction. Can you match it? Please note that capacitor positions C24–C27 must remain empty.

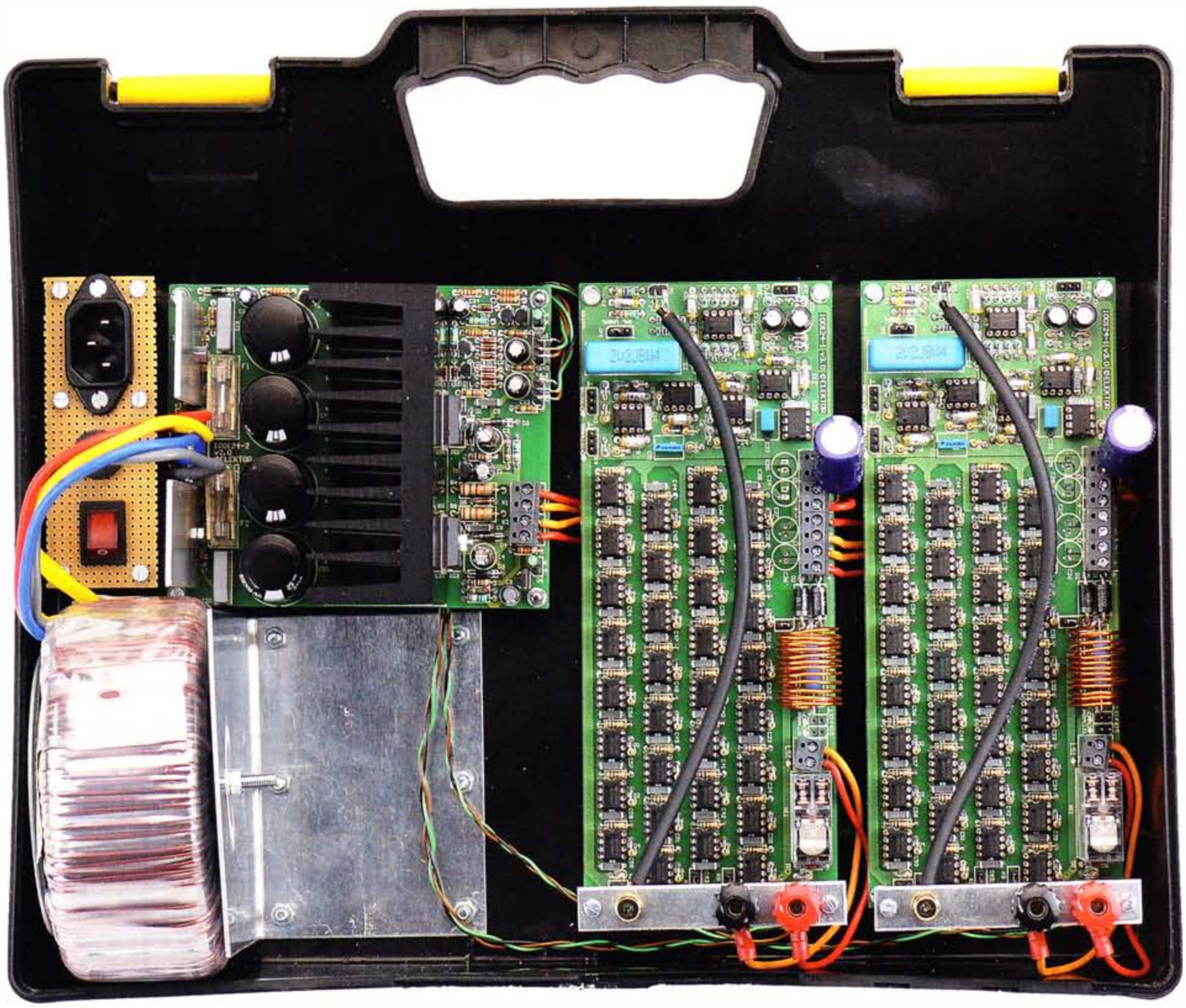


Figure 2. The power supply board is conventional as far as construction and assembly is concerned, but do make sure you get the mounting of the heatsink and the voltage regulators right.
An ABS suitcase as shown here is not recommended as a permanent housing for the amplifier.

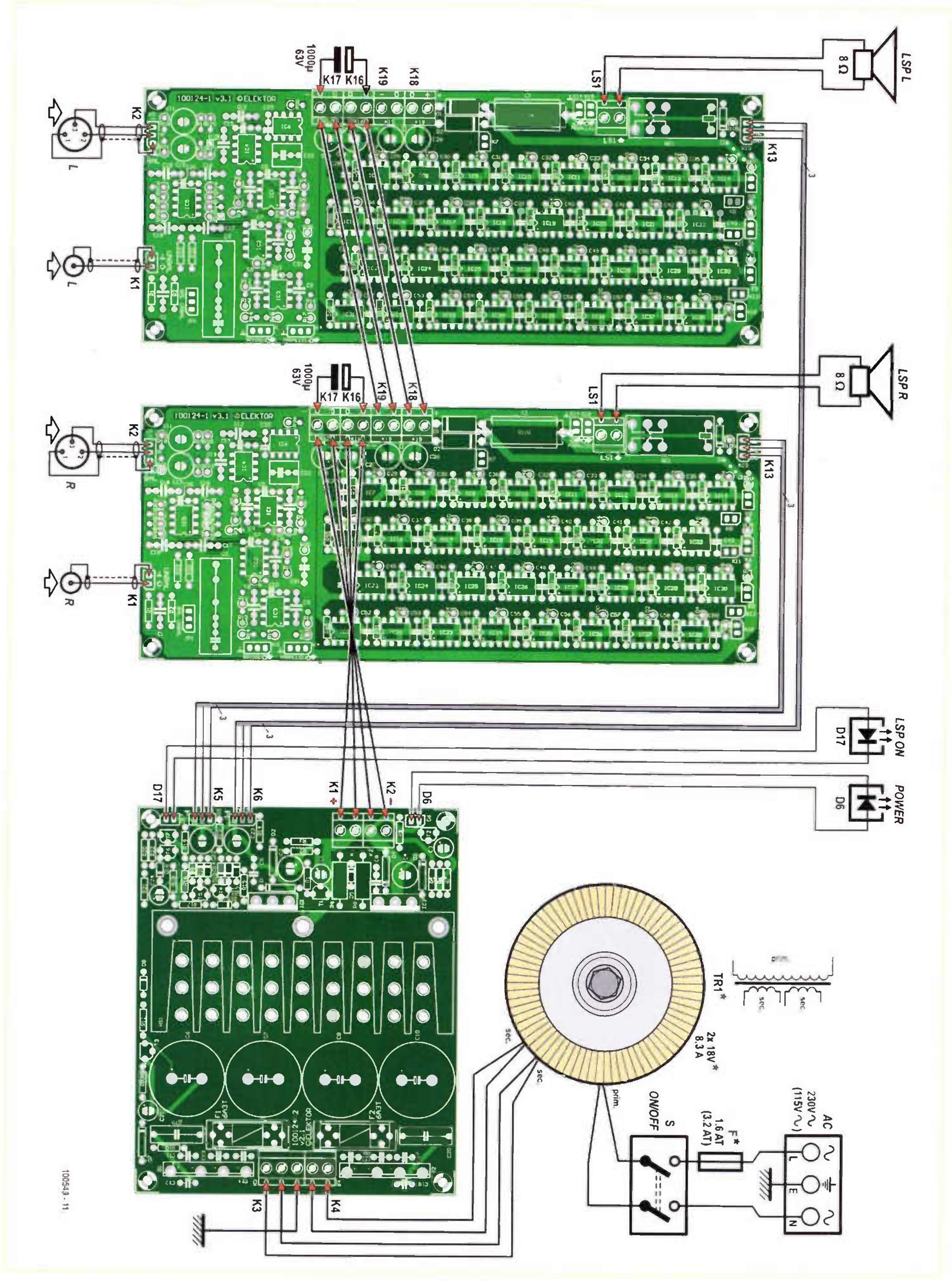


Figure 3. OpAmplifier interwiring diagram. Note that this is for the 2 x 15 watts, 8-ohm version. The electrical ratings of the toroidal transformer and the fuse should match your local AC line voltage. High amps and loudspeaker connections are shown in slightly thicker lines.

Masterclass #1: Extension of design for 4 ohm speakers

It is an inherent property of the 5532 power amplifier that its output current is limited by the internal overload protection of the opamps used. This means that if more current is needed in the load, you need to put more opamps in parallel. The basic design is intended to drive 8 Ohm loudspeakers with a reasonable safety margin, but as it stands it is definitely not recommended for 4 ohm operation.

To extend the 5532 amplifier for 4 ohm operation, it is necessary to double the number of 5532 sections driving the output. This is easier than it might sound because facilities are provided for adding one or more power amplifier PCBs in parallel. The connector K4 (see circuit schematic published last month) is used as an output from the main power amplifier PCB; it comes from the 'zero-impedance' stage IC3A, so the output is at a very low impedance (I measured 0.24 ohms at 1 kHz) but is immune to HF instability caused by cable capacitance. Any desired length of cable can be used. The output from K4 drives an identical power amplifier PCB that has the all the output opamps in place, but the redundant input amplifier circuitry omitted. The equivalent connector K4 on this PCB is used as an input, and drives the output opamps in exactly the same way as on the main power amplifier PCB.

The connectors K14, K15 are used to connect together the outputs of the two amplifier PCBs. Note that this connection is made upstream of the output muting relay RE1, to preserve full protection for the loudspeaker in case of a DC fault.

While this conversion for 4 ohm operation is relatively straightforward, it is essential to remember that the power supply requirements are doubled along with the current output. You will need to consider a larger power transformer, more capable rectifiers, larger reservoir capacitors, and increased supply regulator capacity. In view of these various requirements, this 4 ohm version is mentioned here more as an option for the experimenter rather than a fully cut-and-dried design.

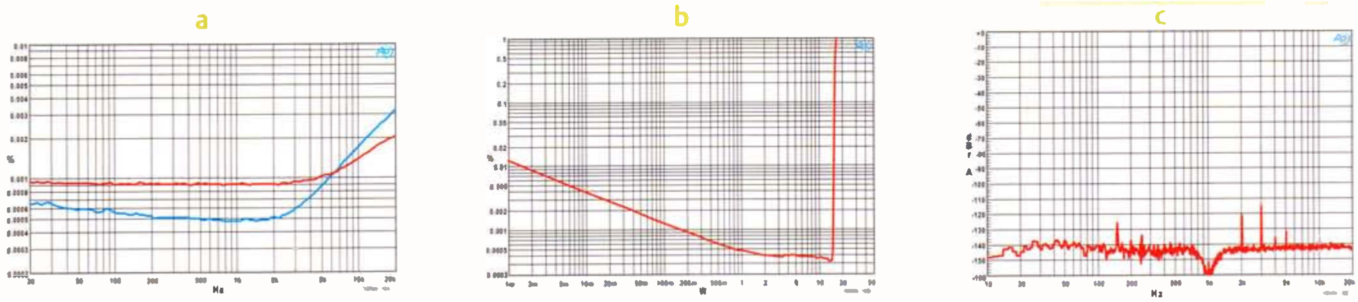


Figure 4. The OpAmplifier was duly 'grilled' on our AP System 2 analyser.
Graph (a): THD & noise vs. output power. Graph (b): distortion vs. output power (1 kHz, BW = 22 KHz).
Graph (c): FFT graph for 1 kHz, 1 watt, 8 ohms.

to assist cooling. The LT1083CP devices are secured to the heatsink with a heat conducting washer inserted. Two holes have to be drilled for the M3 bolts and nuts that hold the regulators firmly on the heatsink. First, determine the location of the holes for the regulators; then secure the heatsink to the board. The regulator terminals should be 'kinked outward' slightly for thermal relief. The two LT1083 regulators should be mounted, secured and soldered last.

Figure 2 shows the PSU board in a "travel & demo" version of the amplifier. PCB standoffs are used as with the amplifier board. The PSU can be tested (carefully) by temporarily connecting the secondaries of the toroidal transformer to the AC input terminals and checking for the correct output voltage of 18.0 VDC ±0.7 V on each rail.

The complete amplifier

The OpAmplifier should be built into in a metal case observing all relevant precautions in respect of electrical safety, specifically with respect to earthing and the use of wiring carrying the AC grid voltage (230 V or 115 V). The size and shape of the case will depend on the number of amplifier boards used, as well as the associated power supply, see the **insets** on **bridged operation** and **4-ohm conversion**. Allow for quite some heat developing in the amplifier case, from the heatsink as well as from the NE5532s. All those milliwatts add up!

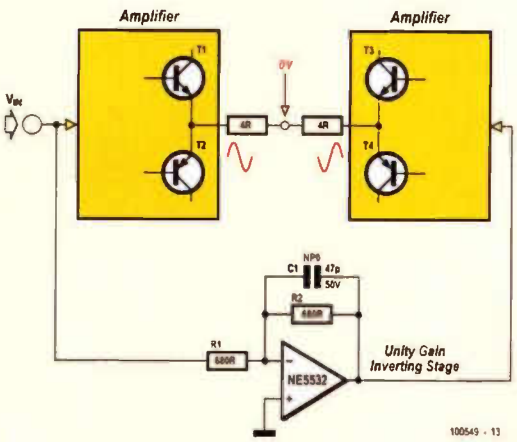
Everything ahead of the bridge rectifier AC terminals should be built, secured and wired with the high currents and voltages always in mind. If possible, do not extend the transformer wires. The amplifier interwiring diagram for the 8 ohm 2 x 15 watts

Masterclass #2: Extension of design for bridged operation

Two power amplifiers are bridged when they are driven with anti-phase signals and the load connected between their outputs; the load is not connected to ground in any way. This method of working doubles the signal voltage across the load, which in theory at least quadruples the output power. It is a convenient and inexpensive way to turn a stereo amplifier into a more powerful mono amplifier. Most conventional power amplifiers do not give anything close to power quadrupling- in reality the increase in available power will be considerably less, due to the power supply sagging and extra voltage losses in the two output stages, which are effectively in series. In most cases you will get something like three times the power rather than four times, and it may be less. I have come across many power amplifiers where the bridged mode only gave twice the output power, and it has to be said that in many cases the bridged mode looks like something of an afterthought.

The 5532 power amplifier will give better results than that for two reasons – firstly the power supply rails are regulated, and will not droop significantly under heavy loading. Secondly, the parallel structure minimises voltage losses; for example, all the 1 ohm output combining resistors are in parallel and their effective resistance is therefore very small.

Bridged mode is so called because if you draw the four output transistors of a conventional amplifier with the load connected between them, as in **Figure A**, it looks something like the four arms of a Wheatstone bridge.



In the drawing, the 8 ohm load has been divided into two 4 ohm halves, to emphasise that the voltage at their centre is zero, and so as far as current output is concerned, both amplifiers are effectively driving 4 Ohms loads to ground. The current capability required is therefore doubled, with all that that implies for increased losses in the output stages. A unity-gain inverting stage is shown generating the anti-phase signal; there are other ways to do it but this is the most straightforward and it simply adds one more 5532 section to a design that already contains quite a few of them. The simple shunt-feedback unity-gain stage shown does the job very nicely, and the 5532 power amplifier incorporates a version of this circuit. The resistors in the inverting stage need to be kept as low in value as possible to reduce their Johnson noise contribution, but not of course made so low that the opamp distortion is increased by driving them. The capacitor C1 across the feedback resistor R2 assures HF stability – with the circuit values shown it gives a roll-off that is –3 dB at 5 MHz, so it does not in any way imbalance the audio frequency response of the two amplifiers.

You sometimes see the glib statement that bridging always reduces the distortion seen across the load because the push-pull action causes cancellation of the distortion products. It is not true. Push-pull systems can cancel even-order distortion products; odd-order harmonics will not be cancelled.

The 5532 power amplifier has facilities for bridging built in. The last stage in the input amplifier is the inverting stage around IC3A, which is needed to get the phase between amplifier input and output correct. If we take the signal off from before this stage, then it is inverted with respect to the amplifier output and can so be used to drive another power amplifier board in anti-phase; this PCB can be built with the input amplifier circuitry omitted, as for the 4 ohm conversion described above. This inverted signal is available at connector K3. The loudspeaker is connected between the two output terminals on the amplifier PCBs and the output ground terminals are not used.

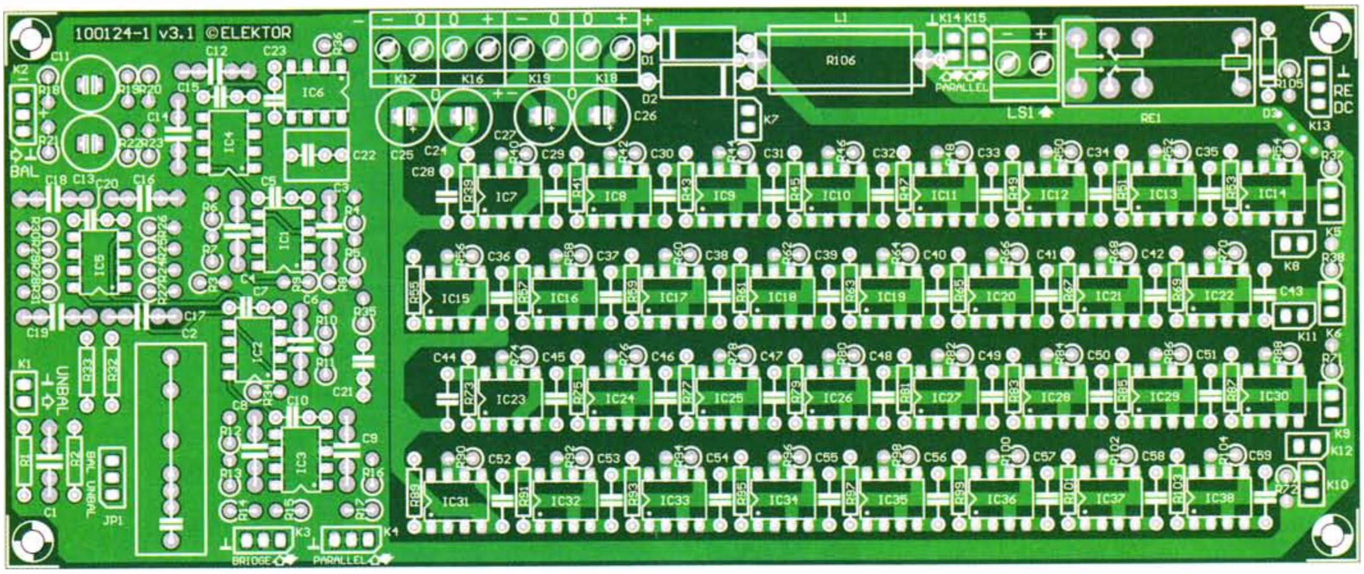
As explained above, driving an 8 ohm load with two bridged power amplifiers means that each amplifier is effectively driving a 4 ohm load to ground, so to take full advantage of the bridging capability requires the output stages to be doubled up, so that there are two power amplifier PCBs in parallel driving in-phase, and two power amplifier PCBs in parallel driving in anti-phase. This is of course quite a serious undertaking, as the number of output opamps has been doubled for bridging, and then doubled again to give adequate output current capability. The power supply capability will also need to be suitably increased.

stereo version of the amplifier is shown in **Figure 3**. The rating of the AC power fuse should match the line voltage in your area (115 V/60 Hz or 230 V/50 Hz). Whichever, an approved IEC style appliance socket with integral fuseholder and double-pole rocker switch must be used.

We should emphasise that the 8.3-amp toroidal transformer indicated in the drawing is advised for the 2 x 15 watt, 8-ohm version of the OpAmplifier. A much beefier transformer is required if you decide to build a 4-ohm or bridged version of the amplifier. As with nearly all audio power amplifiers, it's essential to have a good deal of spare capacity in the 'amps department' so do not skimp on the transformer.

The test results

The basic concept behind the 5532 power amplifier is to create an amplifier which has the very low distortion of the 5532 opamp. Preserving this performance when large currents are flowing to and fro on a circuit board is something of a challenge and requires very careful attention to grounding topology, supply-rail routing, and



COMPONENT LIST

Amplifier board (one channel)

Resistors

5%, 0.25W, Farnell/Multicomp MCF series, unless otherwise indicated.

R1,R14,R17 = 47Ω
R2 = 220kΩ
R3 = 47kΩ
R4,R6 = 910Ω 1% 0.25W, Multicomp type MF25 910R
R5,R7,R10,R11,R15,R16 = 2.2kΩ
R8,R9,R32,R33 = 10Ω
R12 = 1kΩ
R13 = 2.00kΩ 1% 0.25W, Multicomp type MF25 2K
R18,R21 = 150kΩ
R19,R22 = 100Ω
R20,R23 = 68kΩ
R24-R31 = 820Ω
R34,R35 = 10MΩ
R36 = 100kΩ
R37,R38,R71,R72 = 22kΩ
R39-R70,R73-R104 = 1Ω
R105 = 10kΩ
R106 = 10Ω 5% 3W, Tyco Electronics type ROX3SJ10R

Capacitors

C1,C12,C14,C16-C19 = 100pF, 2.5%, 160V, polystyrene, axial (LCR Components)
C2 = 2.2µF 5% 250V, polypropylene, Evox Rifa type PHE426HF7220JR06L2
C3,C4,C6,C8,C9 = 33pF ±1pF 160V, axial, polystyrene (LCR Components)
C5,C7,C10,C15,C20,C23,C28-C59 = 100nF 10% 100V, lead pitch 7.5mm, Epcos type B32560J1104K
C11,C13 = 47µF 20% 35V non polarised, Multicomp type NP35V476M8X11.5 (Farnell # 1236671)
C21 = 22nF 10%, 400V, lead pitch 7.5mm, Epcos type B32560J6223K (Farnell # 9752366)
C22 = 1µF, 10%, 10V, lead pitch 7.5mm, Epcos type B32560J1105K (Farnell # 9752382)
C24-C27 = not fitted, see text

Inductors

L1 = 1.7µH; 10 turns 1 mm (18AWG) ECW, diameter 20mm, effective length 20mm

Semiconductors

D1,D2 = 1N5402
D3 = 1N4148
IC1-IC5,IC7-IC38 = NE5532 (see text for notes on device selection)
IC6 = OP177 (Analog Devices)

Miscellaneous

K1,K5-K12,K14,K15 = 2-pin SIL pinheader, lead pitch 0.1 inch
K2,K3,K4,K13 = 3-pin SIL pinheader, lead pitch 0.1 inch
K16-K19,LS1 = 2-way PCB terminal block, lead pitch 5mm
JP1 = 3-pin SIL pinheader and jumper, lead pitch 0.1 inch
RE1 = PCB mount relay, PCB, DPCO, 24VDC/1100Ω/5A, Omron type G2R-2 24DC
Phono socket, chassis mount, black, Neutrik type NYS367-0
XLR socket, chassis mount, 3-way, Pro Signal type PSG01588
PCB # 100124-1, Elektor Shop, see www.elektor.com/100549

decoupling arrangements. For this reason it is strongly recommended that you use the Elektor PCB design — better still, ready-made boards supplied by Elektor.

The completed prototype was measured using Elektor Labs' **Audio Precision System Two Cascade Plus 2722 Dual Domain** test-set, and **Figures 4a, 4b and 4c** show the pleasing and impressive results. **Figure 4a** shows the harmonic distortion and noise as a function of signal frequency. Measurements carried out at 1 watt (red) and 8 watts (blue) of output power within 80 kHz bandwidth. The graph in **Figure 4b** shows distortion as a function of output power at 1 kHz and a bandwidth of 22 kHz. From about 3 watts the amplifier actually gets close to the lower measurement threshold and noise floor of our analyser! At 15 watts (roughly) the amplifier veers off into clipping. The curve in **Figure 4c**, finally, is an FFT plot for 1 kHz, 1 watt into 8 ohms. The fundamental frequency is suppressed. The second harmonic is way down at –121 dB and the third harmonic slightly higher at –115 dB. At a bandwidth

COMPONENT LIST

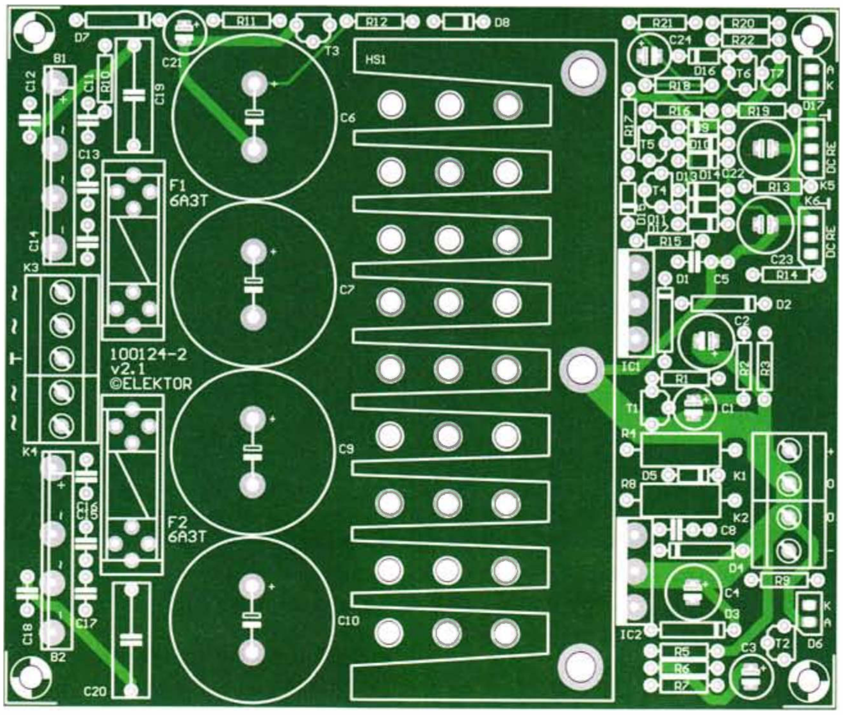
Power supply board

Resistors
5%, 0.25W, Farnell/Multicomp MCF series, unless otherwise indicated.

R1,R5 = 100Ω
R2,R6 = 1.30kΩ 1% 0.25W, Multicomp MF25 1K3
R3,R7 = 39Ω
R4,R8 = 1.5kΩ 5% 1W, Multicomp MCF 1W 1K5
R9,R11,R16,R20-R22 = 2.2kΩ
R10,R12-R15,R19 = 47kΩ
R17 = 150kΩ
R18 = 470kΩ

Capacitors
C1,C3,C24 = 47µF 20% 25V, lead pitch 2.5mm, Rubycon type 25ZLG47M6.3X7
C2,C4 = 100µF 20% 25V, lead pitch 2.5mm, 1.43A AC, Nichicon type UPM1E101MED
C5,C8 = 100nF 10% 100V, lead pitch 7.5mm, Epcos type B32560J1104K
C6,C7,C9,C10 = 4700µF 20% 35V, lead pitch 10mm, snap in, Panasonic type ECOS1VP-472BA (25mm max. diameter)
C11-C18 = 47nF 10% 50V ceramic, lead pitch 5mm
C19,C20 = 47nF 20%, 630VDC X2, lead pitch 15mm, Vishay BCcomponents BFC233620473
C21 = 4.7µF 20% 63V, lead pitch 2.5mm
C22,C23 = 22µF 20% 35V, non polarised, lead pitch 2.5mm, Multicomp NP35V226M6.3X11

Semiconductors
D1-D4,D7 = 1N4002
D5,D8-D16 = 1N4148
D6,D17 = LED, green, 3mm



T1,T3,T4,T6 = BC337
T2,T5,T7 = BC327
IC1,IC2 = LT1083 (Linear Technology)
B1,B2 = GSIB1520 (15A/200V bridge rectifier) (Vishay General Semiconductor)

Miscellaneous
K1,K2,K4 = 2-way PCB terminal block, lead pitch 5mm
K3 = 3-way PCB terminal block, lead pitch 5mm
D6,D17,S1 = 2-pin SIL pinheader, lead pitch 0.1 inch
K5,K6 = 3-pin SIL pinheader, lead pitch 0.1 inch

F1,F2 = fuse, 6.3A antisurge (time lag), with 20x5mm PCB mount fuseholder and cover
Heatsink, Fischer Elektronik type SK92/75SA 1.6 K/W, size: 100x40mm, Farnell # 4621578, Reichelt # V7331G

Miscellaneous, mechanical
M3 screws, nuts and washers for mounting IC1 and IC2 to heatsink (see text)
TO-3P thermal pad, (Bergquist type K6-104)
M4x10 screws for mounting heatsink to PCB
PCB # 100124-2, Elektaor Shop, see www.elektor.com/100549

of 22 kHz, the total level of noise and harmonics is about 0.0005%. If the distortion is measured for the two strongest harmonics only, it is at a level of 0.0002%.

Conclusion

To the non-initiated, the OpAmplifier with its case closed may be just another high-end audio amplifier albeit with quite impressive specs relative to the investment. To the discerning electronic engineer, it's a delightful and off-the beaten path approach to getting high-quality audio watts from a dead common component like the NE5532 you normally associate with nano-ampères and microvolts electronics-wise, and pennies in the financial department!

The proof of the pudding is in the eating. That's why a demo version of the OpAmplifier was cased up in a rugged suitcase for easy transporting, packaging and demoing. At the time of writing, it's about to leave Elektor House on a journey along several audiophiles and audio communities around the world for listening tests. Their feedback is awaited eagerly.

(100549)

Spoilt for choice — which 5532?

It is an unsettling fact that not all 5532 opamps are created equal. The design is made by a number of manufacturers, and there are definite performance differences. While the noise characteristics appear to show little variation in my experience, the distortion performance does appear to vary significantly from one manufacturer to another. Although, to the best of my knowledge, all versions of the 5532 have the same internal circuitry, they are not necessarily made from the same masks, and even if they were, there would inevitably be process variations between manufacturers

Since the distortion performance of the amplifier is unusually, and possibly uniquely, dependent on only one type of semiconductor, it makes sense to use the best parts you can get. In the course of the development of this project 5532s from several sources were tested. I took as wide a range of samples as I could, ranging from brand-new devices to parts over twenty years old, and it was reassuring to find that without exception, every part tested gave the good linearity we expect from a 5532. The information here will be of use not only in the 5532 power amplifier project, but should prove very valuable for anyone using 5532s in high-quality applications.

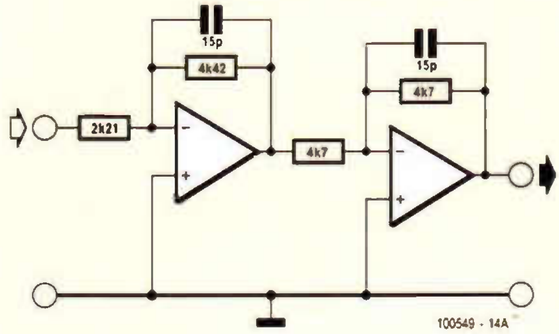
The main sources at present are Texas Instruments, Fairchild Semiconductor, ON Semiconductor, (was Motorola) NJR, (New Japan Radio) and JRC (Japan Radio Company). TI, ON Semi and Fairchild samples were compared in the Elektor labs by Ton Giesberts. The author for his part did THD tests on six samples from Fairchild, JRC, and Texas, plus one old Signetics 5532 for historical interest. The Elektor lab tests were carried out on an actual and very crucial section of the OpAmplifier design: the driver section! See **Figure A** for the circuit and **Figure B** for the AP2 plots.

As it turned out, the Texas 5532s (green line) proved to be distinctly inferior, both in the Elektor Labs and in the author's tests. We have to admit this surprised us, as we have always thought that the Texas part was one of the best available, but the measurements say otherwise. Distortion at 20 kHz ranges from 0.001% to 0.002%, showing more variation than Fairchild and ON Semi as well as being higher in general level. The low-frequency section of the plot, below 10 kHz, is approaching the measurement floor, as for all the other devices, and distortion is only just visible in the noise.

Compared with other maker's parts, the THD above 20 kHz is much higher— and at least 3 times greater at 30 kHz. Fortunately this should have no effect unless you have very high levels of ultrasonic signals that could cause intermodulation. If you have, then you have bigger problems on your hands than picking the best opamp manufacturer...

The graphs show undisputedly that the Fairchild 5532s (blue line) are top of the bill and true audio purists should select these devices even if they come at a slightly higher price as well as being awkward in terms of type code print on the device. Check with your supplier, it should be worth your while. There is effectively no distortion visible above the noise floor up to about 12 kHz, and distortion at 20 kHz is less than 0.0005%.

A



B